6.1200 Quiz 1 — Review Sheet

Wed Oct 14, 7:30 PM · The natural numbers include 0 · appendix = given to you on the quiz · Blue boxes = how to approach a problem

1 · Logic

- **"If A then B"** is false only when A is true and B is false. It means the same as "not A, or B", and the same as its contrapositive "if not B then not A". The converse "if B then A" is a **different** statement.
- **Negating:** "not (A and B)" = "not A or not B". "Not (A or B)" = "not A and not B".
- "Not every x works" = "some x fails". "No x works" = "every x fails".
- **Valid** = true no matter what. **Satisfiable** = true for at least one choice of values.
- Order of "for all" and "there exists" matters: "every person has a mother" ≠ "one person is everyone's mother".

How to approach

1. Are two formulas equivalent? Make a truth table with every row; they're equivalent if the final columns match exactly. If not, name one row where they differ.
2. Translating English: name the domain, then pick the quantifiers in the order the sentence says them.
3. Negating a statement: push the "not" inward one step at a time, flipping for-all ↔ exists and and ↔ or.

2 · Proof methods

- **Direct:** assume the "if" part, work to the "then" part.
- **Contrapositive:** assume the conclusion is false, show the hypothesis is false.
- **Contradiction:** assume the whole claim is false, reach something impossible.
- **Cases:** split up every possibility (e.g. n even / n odd) and prove each.
- **"If and only if":** prove both directions separately.
- **Well-Ordering:** every nonempty set of natural numbers has a smallest element.

How to approach

1. Claim is about every integer and involves even/odd or squares? → **cases by remainder** (n = 2k or 2k+1; or n mod 4).
2. Claim says something is impossible / irrational / doesn't exist? → **contradiction**.
3. "Not B" is easier to work with than "A"? → **contrapositive**.
4. Well-Ordering proof: suppose a counterexample exists, take the **smallest** one, build a smaller counterexample → contradiction.
5. Multi-part question: part (b) almost always uses part (a).

3 · Induction

- **Ordinary:** prove the first case, then prove "if it's true for n, it's true for n+1".
- **Strong:** in the step you may assume it's true for **every** smaller value, not just the one before.
- The predicate P(n) talks about **one** n — never put "for all n" inside it.

How to approach

1. Write "Let P(n) be the statement …".
2. Does the step need the case just before (n−1) only? Ordinary. Does it need one further back (n−2, n/2, …)? Strong.
3. Base cases: prove as many as the step reaches back (step uses n−2 → prove two base cases).
4. Step: "For n ≥ 2, assume P(k) for every k from 0 up to n−1. We show P(n)." Then use the specific earlier case you need.
5. End with "By (strong) induction, P(n) holds for all n."
6. Step won't go through? Try proving something **stronger**.

4 · State machines

- **Invariant:** a property true at the start and kept true by every move — so it's true in every reachable state (the **Invariant Principle**).
- **Derived variable:** a number computed from the state (like b − w, or x + y).
- **Partial correctness:** if the algorithm stops, its answer is right. **Termination:** it does stop. **Total correctness** = both.

How to approach

1. "Can it reach state X?" → expect **no**; look for something X breaks.
2. Find the invariant: for each move, write how each variable changes. Look for a combination that changes by 0 (or by a multiple of some m). Or look for a quantity that only goes up.
3. Check it holds at **every** start state.
4. Check each move type separately preserves it.
5. Show the target state breaks it. Conclude "by the Invariant Principle, the target is unreachable."
6. Termination: find a natural number that goes down on every move.

5 · Sums

- appendix sum of k, sum of k^2, finite geometric sum.
- **Memorize:** sum of $k^3 = (n(n+1)/2)^2$. Infinite geometric $1 + x + x^2 + \ldots = 1/(1-x)$ when $|x| < 1$.

How to approach — pick by the shape of the term

1. **Matches an appendix formula?** Plug in. If the sum starts at m instead of 1: (sum from 1 to n) minus (sum from 1 to m−1). Expand, collect like terms.
2. **A number times a power** (like $k \cdot 2^k$): **perturbation** — write S, write x·S under it shifted one place, subtract, solve for S.
3. **Polynomial in k: ansatz** — the answer is a polynomial one degree higher. Plug in n = 0, 1, 2, … to get equations, solve for the coefficients, prove by induction.
4. **Each term is a difference** f(k+1) − f(k): **telescoping** — everything cancels except the last and first.
5. **Sum inside a sum:** do the inner one first.
6. **No exact formula** (√k, 1/k, log k) or asked for bounds: **integral method**. For an increasing term: sum is between (first term + integral) and (last term + integral). For decreasing: swap first and last.
7. **Always check** your formula with the smallest n by adding the terms by hand.

6 · Asymptotics

- appendix **f is O(g)** means: past some point n_0, f is at most a constant c times g.
- Use the limit of f/g: finite → O; above 0 → Ω (at least as big); finite and above 0 → Θ (same growth); 0 → o (strictly smaller); infinite → ω; exactly 1 → ~ (same including constants).
- **Growth order:** log n < any power n^ε < polynomials < exponentials < n! < n^n.
- appendix **Stirling:** $n! \sim \sqrt{(2\pi n)}(n/e)^n$. Use it whenever n! or a binomial coefficient needs simplifying.
- Θ ignores constant factors; ~ does not. 4^n is not $O(2^n)$ — exponent bases matter.

How to approach

1. "Is f O(g)?" → compute the limit of f/g (use L'Hôpital or divide by the biggest term).
2. "Prove from the definition" → pick a specific c and n_0, then show $f(n) \le c \cdot g(n)$ for all $n \ge n_0$ (bound each term by the biggest one).
3. Ranking functions → take logs to compare, and use the growth order above.

7 · Recurrences

Master Theorem appendix

- For $T(n) = a \cdot T(n/b) + f(n)$, let $c = \log_b a$ and compare f(n) with n^c:
  · f smaller by a power of n → answer n^c
  · f the same as n^c → answer $n^c \log n$
  · f bigger by a power of n, and $a \cdot f(n/b) \le k \cdot f(n)$ for some $k < 1$ → answer f(n)
- **ε** is the size of that power gap (any positive number up to the gap). Write it out, e.g. "$f = n^3 \ge n^{1+\varepsilon}$ with ε = 1".
- Only a log of difference (same power) → no ε exists → theorem doesn't apply. Never use it on T(n−1).

How to approach — pick by the shape

1. **Divides n** (T(n/2)) and they want Θ → **Master Theorem**: find a, b, f; compute c; compare; name the case and ε; check regularity for case 3.
2. **Subtracts, uses only previous terms** ($a_n = 3a_{n-1} + 4a_{n-2}$) → **characteristic equation**: replace a_n by r^2, a_{n-1} by r, a_{n-2} by 1; solve for the roots; answer = (constant)·$(root_1)^n$ + (constant)·$(root_2)^n$; find the constants from the starting values. Repeated root r: use $(A + Bn) \cdot r^n$.
3. **Has an extra constant** (+5): solve without it, then add a constant guess C and solve for C. Constants from starting values come last.
4. **Want an exact answer / anything else → plug and chug**: expand a few times, spot the pattern at step k, go down to the base case, add up, prove by induction.
5. Check your formula against the first few terms.

8 · Divisibility & GCD

- **a divides b** means b = a times some integer. If a divides b and c, it divides any combination sb + tc.
- If a prime divides a product, it divides one of the factors.
- gcd(a, b) = gcd(b, remainder of a ÷ b) = gcd(a − b, b). (Euclid's algorithm.)
- **Bézout:** the gcd can be written as s·a + t·b for some integers s, t. The numbers of that form are exactly the multiples of the gcd.

How to approach

1. Prove something divides something → write each "divides" as "= k times", substitute, factor out.
2. Find a gcd → Euclid: keep replacing the bigger number by the remainder until you get 0.
3. Find s and t (the Pulverizer) → run Euclid; on each line write remainder = x − q·y; substitute earlier lines so every remainder is (something)·a + (something)·b; stop at the gcd. Check the answer.
4. "Is gcd unchanged by this step?" → show any common divisor of the old pair divides the new pair, and vice versa.
5. "Can you make amount m from a and b?" → yes exactly when the gcd divides m.

9 · Modular arithmetic

- **$a \equiv_n b$** means n divides a − b — same remainder when divided by n.
- Rules M1–M7: it acts like "=", and you can add or multiply congruences. So you can **swap any number for its remainder at any time**.
- You can shrink the base of a power, but **not** the exponent mod n. You can cancel a factor k only if gcd(k, n) = 1.
- **Inverse** of a mod n: the number x with $a \cdot x \equiv_n 1$. Exists only if gcd(a, n) = 1.
- appendix **Fermat:** if p is prime and doesn't divide a, then $a^{p-1} \equiv_p 1$. So a^{p-2} is the inverse of a.
- Handy: squares are 0 or 1 mod 4; odd squares are 1 mod 8; a number has the same remainder mod 9 as its digit sum.

How to approach

1. **Remainder of a big power** a^e mod n: (1) shrink a to its remainder (negatives like −1 help); (2) list a, a^2, a^3, … mod n until you hit 1 — call that power k (if n is prime, k = n−1 works by Fermat); (3) every full block of k copies equals 1, so only the leftover matters: divide e by k and use the remainder as the new exponent; (4) if no cycle shows up, square repeatedly (a, a^2, a^4, a^8, …) and multiply the pieces that add up to e.
2. **Find an inverse** → small n: try 1, 2, 3, …; big n: Pulverizer gives $s \cdot a + t \cdot n = 1$, so the inverse is s; prime n: a^{p-2}.
3. **Solve $a \cdot x \equiv_n b$** → multiply both sides by the inverse of a.
4. **Prove something is never true** (e.g. never a square) → check every possible remainder mod a small number (4, 8, 3, 9).

10 · Lessons from the practice quiz

1. **Plug in a small n** to check every formula you derive.
2. **Write the ε** in the Master Theorem, and check regularity in case 3.
3. Claims about every integer → **cases by remainder**, not induction. Use part (a) in part (b).
4. **If ab is odd, then a and b are both odd.**
5. Closing lines: "It holds at every start state and every move preserves it, so by the Invariant Principle…" · "For $n \geq 2$, assume P(k) for all k from 0 to n−1."
6. Unreachable state → look for a quantity that only goes one direction.